Spectroscopy and Atomic Physics

MAIS Physics & Astronomy Club (PAC) · Olympiad lecture + problem handout

8 lecture sections · 15 problems · 1 interactive experiment · 2 vector figures

Start with a physical picture, then introduce only the mathematics needed to make it predictive. The lecture connects everyday reasoning to astronomical measurements. Two vector diagrams and an interactive experiment support the argument; 15 graded problems and independent worked solutions turn the ideas into practice.

▼ Contents & learning path

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1. Atoms leave fingerprints in light

Electrons in atoms occupy quantized energy states. A transition between two states absorbs or emits a photon with

$$\Delta E = hf = \frac{hc}{\lambda}. \tag{1}$$

The same energy gap produces the same rest wavelength whether the line appears in emission or absorption. Spectroscopy turns light into a measurement of composition, motion, temperature, density, magnetic fields and geometry.

Useful lines for the exercises are Hα at 656.28 nm, Hβ at 486.13 nm and Na D near 589.0 nm. Use $c = 299792.458$ km/s and $k_B = 8.617333262 \times 10^{-5}$ eV/K.

2. Three spectra, three geometries

A hot dense source produces an approximate continuum. A thin excited gas emits bright lines. Cooler gas in front of a continuum source removes photons at matching wavelengths, producing absorption lines. The spectrum depends on the source and the material along the line of sight, not only on the chemical element.

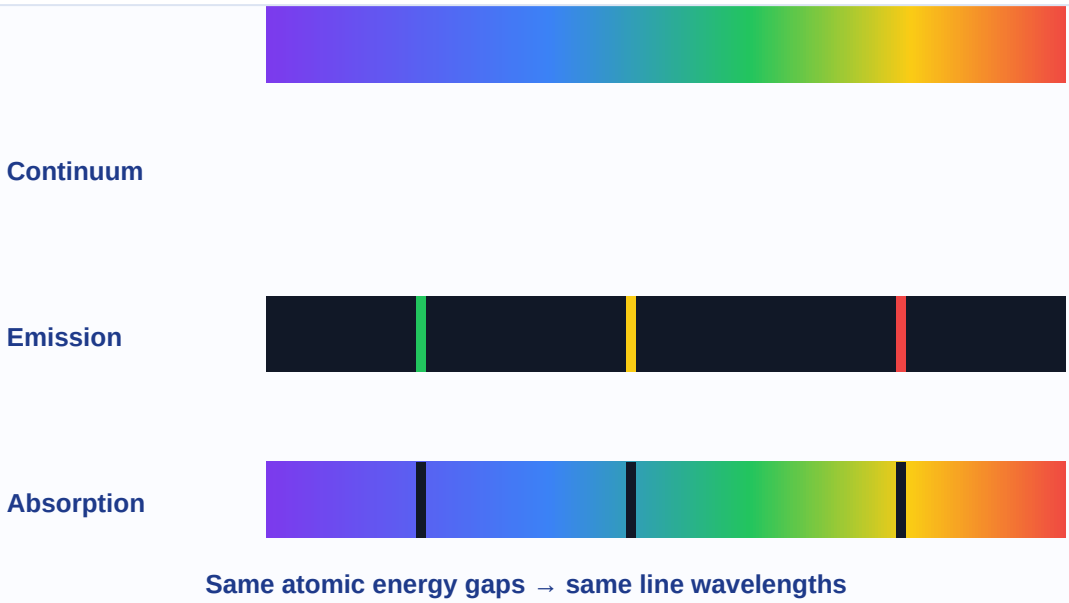


Figure 7. Colours are schematic. Absorption requires a background source and a population able to absorb; it does not mean the atom ceases to exist.

3. A shift measures line-of-sight motion

Define $z = (\lambda_{\rm obs} - \lambda_0)/\lambda_0$. For $|v| \ll c$, $v_r \simeq cz$, with positive velocity meaning recession. For a pure radial special-relativistic Doppler shift,

$$\frac{\lambda_{\rm obs}}{\lambda_0} = \sqrt{\frac{1+\beta}{1-\beta}}, \qquad \beta = v_r/c. \tag{2}$$

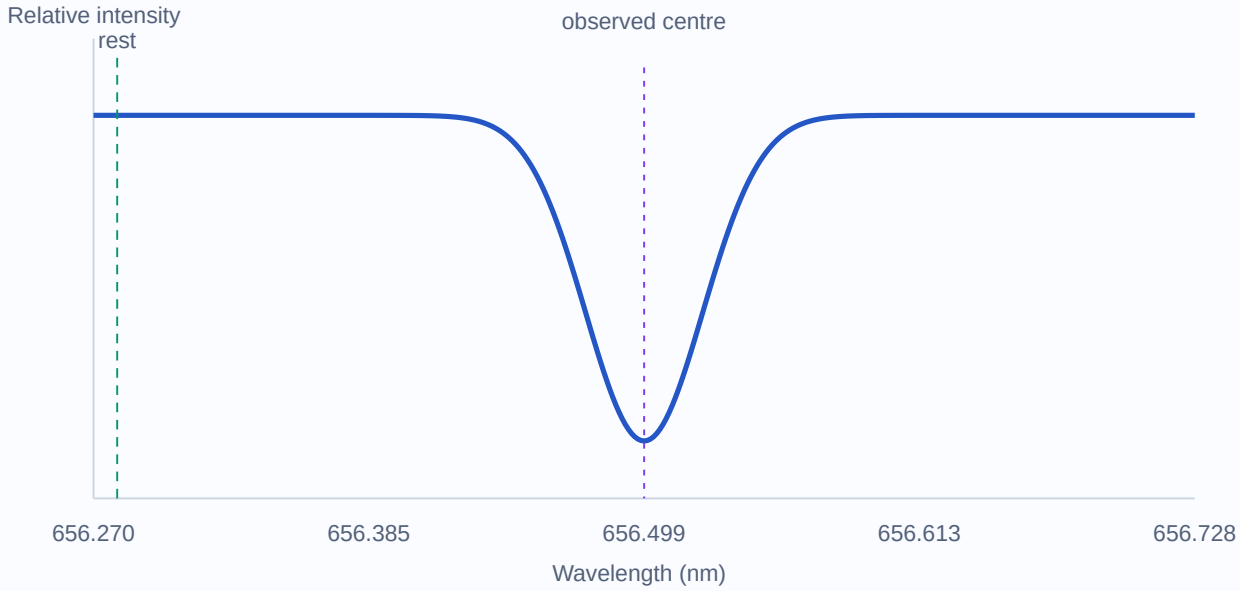
A shift moves the whole pattern. It does not by itself measure transverse motion. Binary orbits, rotation, expansion and gravitational shifts require different interpretations; identify the physical cause before assigning a velocity.

INTERACTIVE EXPERIMENT

Separate a line’s position from its width.

Thermal broadening (H atoms for hydrogen lines, Na atoms for Na D) plus a Gaussian instrumental profile. The relativistic radial Doppler formula sets the centre. Other broadening mechanisms are neglected.

Observed centre **656.4989 nm**	Shift Δλ **0.2189 nm**	Thermal FWHM **0.0468 nm**
Instrument FWHM **0.0328 nm**	Combined FWHM **0.0572 nm**	Thermal σ_v **9.08 km/s**



Thermal broadening uses H atoms for the hydrogen lines and Na atoms for Na D. Gaussian widths combine in quadrature; the instrument does not change the gas temperature.

Try this. Set velocity to zero: the line still has a width. Increase resolving power: does the physical thermal width change? Compare a blue shift with a red shift.

Snapshot of the experiment. Open the HTML edition to change the controls.

4. Width is a different observable

Thermal motion produces a distribution of line-of-sight velocities. For atoms of mass m,

$$\sigma_v = \sqrt{\frac{k_B T}{m}}, \qquad \sigma_\lambda = \lambda \frac{\sigma_v}{c}, \qquad \Delta\lambda_{\mathrm{FWHM}} = 2\sqrt{2\ln 2}\,\sigma_\lambda. \tag{3}$$

The instrument also broadens the line. Resolving power $R = \lambda/\Delta\lambda$ describes an instrumental width. Independent Gaussian contributions add in variance, so their FWHMs add in quadrature. Natural, pressure,

rotational and bulk-flow broadening need not be Gaussian.

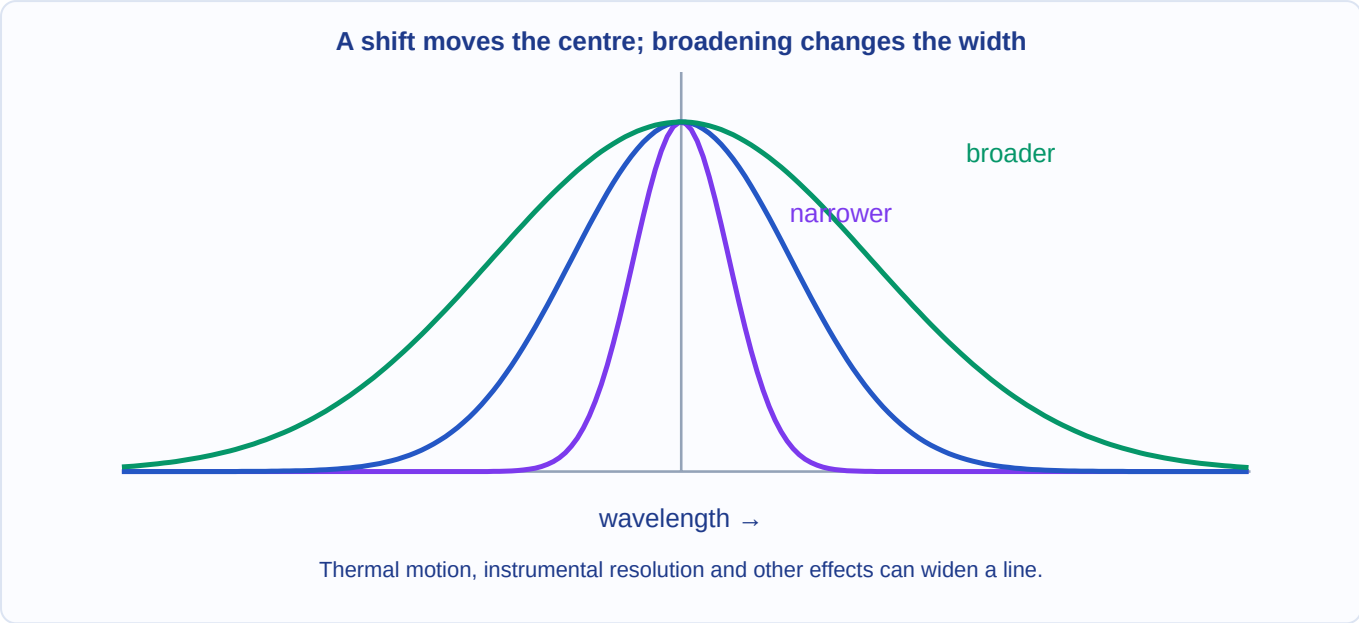


Figure 8. These are schematic emission profiles. For independent Gaussian contributions, variances add; full widths at half maximum add in quadrature only when all contributions are Gaussian.

5. Line strength and optical depth

A line's centre can be shifted while its equivalent width stays nearly fixed. Equivalent width measures the area removed from, or added to, the continuum in wavelength units. In the optically thin limit it grows approximately with the number of absorbers. Saturated lines grow more slowly; strong damping wings change the growth again.

For pure absorption through a uniform screen,

$$I = I_0 e^{-\tau}. \tag{4}$$

Optical depth $\tau = 1$ leaves $1/e$ of the background intensity. A deeper line does not always mean a proportionally larger abundance.

6. Splitting: magnetic fields and level populations

A magnetic field can split a line. In the normal Zeeman effect, the frequency shift of each shifted component is

$$\Delta\nu = \frac{eB}{4\pi m_e}. \tag{5}$$

Real atoms can show more complicated anomalous patterns. Level populations also matter: in thermal equilibrium, $n_2/n_1 = (g_2/g_1)e^{-\Delta E/(k_B T)}$. A line can be weak because few atoms occupy the relevant lower level, not because the element is absent.

7. Polarisation and scattering

Polarisation describes the orientation of the electric-field oscillation. An ideal polarizer transmits $I = I_0 \cos^2 \theta$ from already polarized light (Malus's law). Unpolarized light loses half its intensity through the first ideal polarizer. Scattering can polarize light, especially near a 90° scattering angle. Reflection at Brewster's angle, $\tan \theta_B = n_2/n_1$, is fully polarized in the ideal dielectric case.

8. Build an interpretation from several clues

Use line identification for composition, centroid for bulk motion, width for thermal/instrumental/dynamical effects, depth and equivalent width for populations and radiative transfer, splitting for magnetic fields, and polarization for geometry. No single number tells the whole story.

Check before claiming. Subtract the instrumental width before inferring temperature. Correct velocities to a stated reference frame. Distinguish a line shift from a line width, and a wavelength shift from a change in the photon's speed in vacuum.

9. Lecture + Problems: predict, calculate, check

Fifteen original problems: **[1]** foundation, **[2]** multi-step Olympiad reasoning, **[3]** challenge. Use the constants supplied in the lecture unless a question overrides them. Explain assumptions and keep units. Every problem has a worked solution.

[1] Problem 1 — A small redshift. Hα at 656.28 nm comes from a source receding at 100 km/s. Find the observed wavelength using the relativistic radial formula and compare with cz.

Worked solution →

[2] Problem 2 — Two stars in one spectrum. A 500.0 nm rest line is observed at 499.8 and 500.3 nm from the two stars in a circular edge-on binary observed at quadrature. Find both radial velocities, the systemic velocity and the mass ratio.

Worked solution →

[3] Problem 3 — Weighing a rotating galaxy. A galaxy's systemic velocity is 5000 km/s. Opposite sides along its major axis differ by ±200 km/s. The inclination is 30°. Find the deprojected rotation speed and enclosed mass at radius 5 kpc, assuming circular rotation.

Worked solution →

[2] Problem 4 — Thermal width. Hydrogen gas at 10,000 K produces Hα. Use $m_H = 1.00784u$. Find the one-dimensional thermal velocity dispersion and the thermal FWHM in km/s and nm.

Worked solution →

[1] Problem 5 — Can the spectrograph separate them? A spectrograph has $R = 20000$ at 500 nm. Find its instrumental FWHM in nm and km/s. Can it resolve a doublet separated by 0.040 nm?

Worked solution →

[3] Problem 6 — A finite lifetime. An excited state lives for 10 ns. Using $\Delta E \Delta t \sim \hbar/2$, estimate the natural frequency width and wavelength width near 500 nm. State the convention-dependent nature of the estimate.

Worked solution →

[3] Problem 7 — A magnetic fingerprint. For a normal Zeeman pattern in a 0.1 T field, find the frequency shift of each shifted component and its wavelength shift near 500 nm. What is the separation between the two shifted components?

Worked solution →

[1] Problem 8 — Two polarizers. Fully polarized light passes through an analyzer at 60° to its polarization. What fraction emerges? Repeat for initially unpolarized light passing through two ideal polarizers with the same relative angle.

Worked solution →

[2] Problem 9 — Brewster's angle. Light travels from air into glass of refractive index 1.5. Find Brewster's angle and describe the reflected polarization in the ideal case.

Worked solution →

[2] Problem 10 — More atoms, not always twice the depth. Explain how equivalent width grows with absorber number in the optically thin, saturated and damping-wing regimes. Why is line depth alone a poor abundance measure?

Worked solution →

[1] Problem 11 — How transparent? For pure absorption, find transmitted fractions at $\tau = 0.2$ and 2. Find the optical depth that halves the intensity.

Worked solution →

[2] Problem 12 — Populating an excited level. Two atomic levels have equal statistical weights and separation 2 eV. Find n_2/n_1 at 6000 and 12,000 K in thermal equilibrium.

Worked solution →

[1] Problem 13 — Why blue scatters more. In the Rayleigh limit, scattering strength scales as λ^{-4}. Compare 400 and 800 nm. At what scattering angle is single-scattered polarization strongest in the ideal case?

Worked solution →

[3] Problem 14 — Expansion parallax. A spherical shell expands at 100 km/s. Its angular radius is 1 arcsecond and is observed to grow at 0.10 arcsecond per year. Assuming the spectroscopic expansion speed also equals the transverse speed of the visible rim, find the distance from the rim's proper motion. How much does the radius grow in 10 years?

Worked solution →

[3] Problem 15 — An unresolved line through a spectrograph. A Gaussian absorption line has intrinsic depth 0.4 and FWHM 0.10 nm. Find its equivalent width. A Gaussian instrument response has the same FWHM. Find the observed FWHM and peak depth, assuming equivalent width is conserved.

Worked solution →

Further practice and provenance

The 15 exercises above and their solutions are **original MAIS PAC problems**, inspired by the skills in this syllabus topic. They are not copied from the papers below. The earlier club lecture compiled these official past-paper identifiers for further practice; only identifiers are reproduced, not copyrighted question statements or official solutions. Round and question labels retain the earlier compilation's notation and should be checked against the official paper.

- IOAA 2007 · TH · 1.8
- IOAA 2007 · TH · 1.12
- IOAA 2008 · TH · L2
- IOAA 2009 · TH · P2

- IOAA 2012 · TH · S9
- IOAA 2013 · TH · S13
- IOAA 2014 · TH · P7
- IOAA 2014 · TH · P8
- IOAA 2015 · TH · S6
- IOAA 2015 · TH · S8
- IOAA 2015 · DA · D2
- IOAA 2016 · TH · T8
- IOAA 2017 · TH · T4
- IOAA 2018 · TH · T6
- IOAA 2019 · TH · 5
- IOAA 2021 · TH · Q4
- IOAA 2023 · TH · Q2
- IOAA 2025 · TH · T04

Sources and scope. Topic coverage follows the IOAA syllabus. Official problems and constants are available through the IOAA past-paper archive. The derivations, diagrams, experiments and numerical teaching examples in this handout were prepared independently for MAIS PAC. Calculator-level methods are emphasized; optional calculus connections are explicitly identified.